

Finding a Fraction of a Number

Answer Key

Grade 4–5

Quick Answers

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| <p>1. 3</p> <p>2. 6</p> <p>3. (a) one third of 18 = 6, (b) two thirds of 18 = 12</p> <p>4. 12</p> <p>5. 7</p> | <p>6. (a) one eighth of the class = 3, (b) three eighths of the class = 9</p> <p>7. 24</p> <p>8. 35</p> <p>9. 12.5 dollars</p> <p>10. (a) number of seats in the theater = 28, —</p> |
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What is verified

9 of 10 problems fully machine-checked against SymPy.
 — marks an answer only you can judge — problem 10. The worked solution states what a correct response must contain.

Curriculum

Level: Grade 4–5

5.NF.B – problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10

Difficulty 1–5 of 5 across 13 tagged checks.

- 1.** The picture shows 12 counters. Find $\frac{1}{4}$ of 12.

Solution: “ $\frac{1}{4}$ of 12” means split 12 counters into 4 equal groups and take one group.
 $12 \div 4 = 3$, so each group holds 3

- 2.** Find $\frac{1}{5}$ of 30.

Solution: Split 30 into 5 equal groups: $30 \div 5 = 6$. One group is 6

- 3.** Use equal groups to find two thirds of 18: (a) $\frac{1}{3}$ of 18, (b) $\frac{2}{3}$ of 18.

Solution: (a) Split 18 into 3 equal groups: $18 \div 3 = 6$, so $\frac{1}{3}$ of 18 is 6

(b) Two thirds is two of those groups: $2 \times 6 = 12$, so $\frac{2}{3}$ of 18 is 12

- 4.** Kofi gives $\frac{3}{5}$ of his 20 stickers to his sister. How many does he give away?

Solution: First one fifth: $20 \div 5 = 4$ stickers. Then three fifths is three groups of 4:
 $3 \times 4 = 12$. He gives away 12 stickers.

- 5.** A book has 56 pages; Lena reads $\frac{1}{8}$ of it. How many pages?

Solution: One eighth of 56 is $56 \div 8 = 7$. She reads
pages.

7

6. 24 students; $\frac{3}{8}$ of the class walks to school. (a) students in $\frac{1}{8}$ of the class; (b) students who walk.

Solution: (a) One eighth of the class: $24 \div 8 = 3$ students

3

(b) Three eighths is three groups of 3: $3 \times 3 = 9$ students walk

9

7. One fourth of Maya's marbles is 6. How many marbles in all?

Solution: Work backward from one group to the whole. If one of the 4 equal groups holds 6 marbles, the whole is $4 \times 6 = 24$. Maya has
marbles. *Check:* $\frac{1}{4}$ of 24 = 6. ✓

24

8. 42 apples; $\frac{5}{6}$ are red. How many red apples?

Solution: One sixth of 42: $42 \div 6 = 7$. Five sixths is five groups of 7: $5 \times 7 = 35$. There
are
red apples.

35

9. A gift card holds 20 dollars; Nia spends $\frac{5}{8}$ of it. How many dollars does she spend?

Solution: One eighth of 20 dollars: $20 \div 8 = 2.50$ dollars. Five eighths is $5 \times 2.50 = 12.50$.
She spends 12.50 dollars

10. Challenge: $\frac{3}{4}$ of the seats are filled, and 21 seats are filled. (a) How many seats? (b) Explain with a drawing of equal groups.

Solution: (a) The 21 filled seats make 3 of the 4 equal groups, so one group (one fourth) is $21 \div 3 = 7$ seats. The whole is $4 \times 7 = 28$. The theater has
seats.

28

Explanation (instructor judges): a correct drawing or sentence shows 4 equal groups of 7, with 3 groups (21 seats) filled — so one fourth is 7 and the whole is 28.